

File Handling with PHP



Files and PHP

- File Handling
 - Data Storage
 - Though slower than a database
 - Manipulating uploaded files
 - From forms
 - Creating Files for download



Open/Close a File

- A file is opened with `fopen()` as a “stream”, and PHP returns a ‘handle’ to the file that can be used to reference the open file in other functions.
- Each file is opened in a particular **mode**.
- A file is closed with `fclose()` or when your script ends.



File Open Modes

<code>'r'</code>	Open for reading only. Start at beginning of file.
<code>'r+'</code>	Open for reading and writing. Start at beginning of file.
<code>'w'</code>	Open for writing only. Remove all previous content, if file doesn't exist, create it.
<code>'a'</code>	Open writing, but start at END of current content.
<code>'a+'</code>	Open for reading and writing, start at END and create file if necessary.



File Open/Close Example

```
<?php
// open file to read
$storead = fopen('some/file.ext', 'r');
// open (possibly new) file to write
$towrite = fopen('some/file.ext', 'w');
// close both files
fclose($storead);
fclose($towrite);
?>
```



Now what..?

- If you open a file to read, you can use more in-built PHP functions to read data..
- If you open the file to write, you can use more in-built PHP functions to write..



Reading Data

- There are two main functions to read data:
- **fgets (\$handle, \$bytes)**
 - Reads up to \$bytes of data, stops at newline or end of file (EOF)
- **fread (\$handle, \$bytes)**
 - Reads up to \$bytes of data, stops at EOF.



Reading Data

- We need to be aware of the End Of File (EOF) point..
- **feof(\$handle)**
 - Whether the file has reached the EOF point. Returns true if have reached EOF.



Data Reading Example

```
$handle = fopen('people.txt', 'r');  
  
while (!feof($handle)) {  
    echo fgets($handle, 1024);  
    echo '<br />';  
}  
  
fclose($handle);
```



Data Reading Example

```
$handle = fopen('people.txt', 'r');
```

```
while (!feof($handle)) {  
    echo fgets($handle, 1024);  
    echo '<br />';  
}
```

Open the file and assign the resource to \$handle

```
fclose($handle);
```



Data Reading Example

```
$handle = fopen('people.txt', 'r');
```

```
while (!feof($handle)) {  
    echo fgets($handle, 1024);  
    echo '<br />';  
}
```

```
fclose($handle);
```

While NOT at the end of the file, pointed to by \$handle, get and echo the data line by line



Data Reading Example

```
$handle = fopen('people.txt', 'r');  
  
while (!feof($handle)) {  
    echo fgets($handle, 1024);  
    echo '<br />';  
}
```

```
fclose($handle);
```

Close the file



File Open shortcuts..

- There are two 'shortcut' functions that don't require a file to be opened:
- `$lines = file($filename)`
 - Reads entire file into an array with each line a separate entry in the array.
- `$str = file_get_contents($filename)`
 - Reads entire file into a single string.



Writing Data

- To write data to a file use:
- **`fwrite($handle, $data)`**
 - Write \$data to the file.



Data Writing Example

```
$handle = fopen('people.txt', 'a');
```

```
fwrite($handle, "\nFred:Male");
```

```
fclose($handle);
```



Data Writing Example

Open file to append data (mode 'a')

```
$handle = fopen('people.txt', 'a');
```

```
fwrite($handle, "\nFred:Male");
```

```
fclose($handle);
```

Write new data (with line break after previous data)



Other File Operations

- Delete file
 - `unlink ( 'filename' ) ;`
- Rename (file or directory)
 - `rename ( 'old name' , 'new name' ) ;`
- Copy file
 - `copy ( 'source' , 'destination' ) ;`
- And many, many more!
 - www.php.net/manual/en/ref.filesystem.php



Dealing With Directories

- Open a directory
 - `$handle = opendir('dirname');`
 - `$handle` 'points' to the directory
- Read contents of directory
 - `readdir($handle)`
 - Returns name of next file in directory
 - Files are sorted as on filesystem
- Close a directory
 - `closedir($handle)`
 - Closes directory 'stream'



Directory Example

```
$handle = opendir ( './' );  
  
while ( false !== ( $file = readdir ( $handle ) ) )  
{  
    echo "$file<br />";  
}  
  
closedir ( $handle );
```



Directory Example

```
$handle = opendir('.') ;
```

Open current directory

```
while(false !== ($file=readdir($handle)))  
{  
    echo "$file<br />";  
}
```

```
closedir($handle) ;
```



Directory Example

```
$handle = opendir(' ./ ' );
```

```
while( false !== ( $file = readdir( $handle ) ) )  
{  
    echo "$file<br />";  
}
```

```
closedir( $handle );
```

Whilst readdir() returns a name, loop through directory contents, echoing results



Directory Example

```
$handle = opendir(' ./ ' );  
  
while( false !== ( $file = readdir( $handle ) ) )  
{  
    echo "$file<br />";  
}
```

```
closedir( $handle );
```

Close the directory stream



Other Directory Operations

- Get current directory
 - `getcwd()`
- Change Directory
 - `chdir('dirname');`
- Create directory
 - `mkdir('dirname');`
- Delete directory (MUST be empty)
 - `rmdir('dirname');`
- And more!
 - www.php.net/manual/en/ref.dir.php



Review

- Can open and close files.
- Can read a file line by line or all at one go.
- Can write to files.
- Can open and cycle through the files in a directory.

